Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping **JavaScript**, **Node.js**, and React products with Python **computer vision** and ML pipelines, Firebase and GCP backends, Docker, REST APIs, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Advanced NLP, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built ingestion and integration pipelines across PostgreSQL, SQL Server, and **cloud storage** with incremental loads, validation, and CI/CD-backed releases, sustaining 99%+ reliability over 5K+ monthly production jobs.
- Tuned queries and access patterns for a 500+ user platform, cutting dashboard latency from 1.2s to 350ms, and restructured async AI job processing to **triple throughput** across 1K+ jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built an end-to-end product with a React upload and review interface and Python APIs, routing handwritten **image** submissions through GPT-4o Vision and Claude Vision OCR for 150+ instructors.
- Designed concurrent batch processing for 1K+ uploaded files per run with secure PDF, DOCX, and ZIP ingestion, handling 10K+ **daily requests** and cutting instructor feedback turnaround 60% in production.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Automated internal processes** with nightly Jenkins jobs that built executables, installed them on test VMs, ran 20+ unit and integration suites, and emailed reports, adding testing rigor across releases.
- Integrated large-scale third-party **APIs** and optimized them with pagination, field filtering, and caching, improving response times 30% while redesigning AWS Glue pipelines for 68% higher pipeline efficiency in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **Node.js** and React platform with RESTful microservices and real-time WebSocket updates for resource allocation, billing, and leads, syncing 2M+ records across multiple locations with consistent real-time state.
- Wrote Jest and Selenium **tests** gating Jenkins CI/CD with Docker packaging, and used Google Analytics on drop-off points to streamline flows, cutting resource wastage 42% and lifting conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built APIs on **Firebase** with Firestore real-time sync, image storage, and Cloud Messaging triggers for a React Native and React booking app, speeding booking confirmations 40% across 50K+ records.
- Integrated external payment providers through idempotent webhooks for refunds and disputes, cutting manual handling about 70%, and presented the end-to-end architecture to the client before any implementation work began.

Skills

Languages: JavaScript, TypeScript, Python, SQL, Java, HTML, CSS
Web: Node.js, Express.js, React, Next.js, React Native, REST APIs, WebSockets, Webhooks
Computer Vision & ML: PyTorch, TensorFlow, Transformers, CLIP, SAM, OCR, Stable Diffusion
Cloud & Data: Firebase, Firestore, GCP, AWS, Docker, PostgreSQL, MongoDB, FAISS, Linux
Quality & Tools: Jest, Selenium, PyTest, Unit Testing, CI/CD, GitHub Actions, Jenkins, Git, Google Analytics

Projects

Polygon Annotation Editor | *React, TypeScript, Canvas, Node.js, Firebase, Jest*

- Designed a browser **annotation** tool for drawing boxes and polygon segmentation masks on images, with undo history, keyboard shortcuts, and model-assisted suggestions that a labeler can accept or adjust.

Image Dataset Search and Filtering | *Node.js, React, Firestore, Python, CLIP*

- Designed a **filtering** interface that lets users slice large image datasets by capture time, GPS location, custom tags, and CLIP text search, backed by indexed metadata queries for fast results.

PiXMod Live | *Next.js, SAM, Stable Diffusion, CLIP, MongoDB, Stripe*

- Built a **computer vision** SaaS with SAM image segmentation, Stable Diffusion inpainting and restoration, and CLIP search over 100K+ indexed image embeddings, processing 1K+ edited assets in MongoDB.